



Unit 7: Resource Management

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Introduction to the Unit

In this unit, we will learn about the concept of *Resource Management*. Upon completion of this unit, we will have a fair understanding of the '*people*' aspect of project management.

We will focus on understanding the importance of *People Management* and conducting *Team Appraisals* and *One-on-Ones*. We will also understand the various *Resource Considerations* in a *Project* and will conclude the unit by understanding how to *Manage* and *Motivate* the *Team Members*.

Objectives

Upon completion of this unit, you will be able to:

- Explain the concept of people management
- Describe the best practices in conducting team appraisals and one-on-ones
- Explain how to meet the resource needs of a project
- Describe the process of assigning work to resources
- Identify the resource considerations in a project
- Explain how to manage resources and keep them motivated



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1. Introduction
2. Project Management is People Management
3. Team Appraisals
4. One-On-Ones
5. Summary

Objectives

Upon completion of this segment, you will be able to:

- Explain the importance of people management
- Describe the process of conducting team appraisals
- Describe the process of conducting one-on-ones (meetings with individual team members)

Introduction

In the last segment, we learned about contract management, the various stages of contracting and the different types of contracts. In this segment, we will focus on understanding the importance of people management. We will also discuss the process of conducting team appraisals and one-on-ones with *Team Members*.

Project Management is People Management

The key to delivering successful projects, consistently, is people. This is so because the strength of the project management framework and tools and the completeness of documentation is directly dependent on the project team.

Why focus on people?

- It fosters professionalism in project management
- It is legally binding to protect the well-being of employees
- It helps in risk management
- It is a prerequisite for success in a globalised economy
- It makes for good business sense as well as common sense

While relating to the above reasons, our focus should be upon culture and diversity. As a professional body of *Project Managers* (risk managers), we need to realise the commonalities shared by different individuals along with their set of differing world view in order to galvanise the team towards success. We have ample instances of failure in ensuring risk management, most of which could have been avoided if people were given the importance they deserve in a project.



Investing in people

The role of the *Project Manager* is ever-changing—our responsibilities and teams, often change shape frequently during a project life cycle. An effective Project Manager recognises the rewards of investing in his team members.

While technical skills are deployed and enhanced in a linear fashion during the life of a project, people skills—in marked contrast—are employed throughout the project in all phases. This alone is a sufficient reason for giving due importance to people skills, given their critical role in the overall success of the project.

People skills need to be thoroughly integrated into the project development and the team building process. These skills need to be maintained consistently throughout all the phases of a project. Our ability to relate to individuals, to create teams, to elicit the best from project contributors and influence decision-making, throughout the project life cycles and beyond, is the essence of good project management.

People first, always

The opportunity to view people as unique individuals rather than as a consolidated group that we may like/do not like or appreciate/detest enables greater learning and openness to share knowledge and experience. As project contributors, managers, *Sponsors* and other *Stakeholders*, we have greater effectiveness regardless of how brief or protracted the communication exchanges may be.

As project managers, we can increase the overall effectiveness of our projects and the organisations for which we work by selecting our *Team Members* with more awareness. This may involve stepping out of our prejudices that recognising that age is not necessarily a guarantee of rich experience and youth is not necessarily a reliable predictor unreliability or otherwise.

To be effective and successful in putting people first, a *Project Manager* must recognise that he/she is also a person with a set of values, beliefs and experiences. Since we are constantly evolving and growing, we have the opportunity to shape our communication much more wisely and more successfully if we realise that people are our greatest resource and that they have the potential of adding much value to a project.

Categorising people – with moderation

Categorising our lives gives us a sense of perspective and allows us to move through our daily lives quickly and easily. Checking the weather forecast to determine whether to carry an umbrella, choosing the best time of day for a meeting with a project *Sponsor*, or which day of the week to hold a team building event is all about making sound choices when it comes to dealing with people.

Categorising people too readily under the influence of prejudice is inefficient and can be counter productive. Assessing the suitability of an individual, group or corporation to provide a service,



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partnership or skill set based on a range of interactions is a positive way forward to good decision making in choosing the appropriate team members who will contribute to the success of the project.

However, letting our prejudices come in our way of categorising people effectively will foster bias and do injustice to the project.

There is an anecdote that is often quoted to highlight the existence of prejudices in everyone's minds and how to deal with them. As per this account, there is a certain museum where the curator takes the visitors on a guided tour of the museum. At one point during the tour, the group of visitors come across a closed door which has a sign that reads 'enter only if you do not have any prejudices'.

Invariably, there is always an overenthusiastic person in the group who steps forward, announces that he/she carries no prejudices and opens the door only to be humbled on finding a brick wall staring him/her in the face.

At this point, the curator steps forward to explain very gently that the purpose of having the sign on the door is to make us realise that all of us have prejudices as we grew up with our unique set of life experiences, which conditioned us to have prejudices and biases.

Therefore, we must not deny the existence of prejudices, rather we should be aware of them and ensure that we do not let them cloud our vision and affect our judgement while dealing with others.

This anecdote has much relevance to any activity involving association with others, including project management where a *Project Manager* will have his/her prejudices but will need to ensure that they do not come in the way of building honest and fair relationships with the *Team Members*.

Increasing awareness and, therefore, understanding an individual more completely, enables people management, reduces negative risk and increases project success.

Celebrating diversity

A *Project Manager* must realise that very rarely will there be an opportunity where there is absolute homogeneity in the team composition in terms of capabilities, cultural background, life experiences and the little quirks and fancies that differentiate one *Team Member* from the other.

Therefore, it is futile to make an attempt to standardise the team composition by having only those employees who are similar in every respect. Moreover, a project will only benefit from the diverse experiences and viewpoints of people from different walks of life.

It makes sense for a *Project Manager* to continue to celebrate the diversity and uniqueness in his/her team while appreciating the close similarities.

Project Management, at the end of the day, is after all, *People Management*!



Performance Appraisals

The fear of appraisals!

There is an anecdote about a man who owned a poultry farm. He was able to earn a decent amount by selling eggs and other poultry products. One fine day, someone suggested to him to enrol for a management program where he would learn many strategies for increasing his profits as an entrepreneur.

The man took the advice and handed over his poultry farm to an associate for a couple of years while he took a sabbatical to enhance his professional qualifications. When he returned after completing his management degree, he decided to operate his poultry farm in a corporate manner and called for a kick-off meeting where he addressed his flock of hens and informed them about their Key Result Area (KRA) of laying at least two eggs in a day.

The hens were informed that the team performance would be evaluated in the appraisal the next day. The next day, every hen lay at least two eggs. The man was pleased but noticed that one of the hens had laid only one egg. He called for a one-to-one meeting with the hen and asked her why she had not taken the appraisal seriously and had laid only one egg despite knowing that laying two eggs a day was her primary KRA.

The hen apologised and went on to explain that she had not taken the appraisal lightly, rather it was because of her fear of the appraisal that she had been able to lay one egg despite being a rooster!

This story mirrors the actual feelings and anxieties of many employees who are facing an impending appraisal. Our life experiences have taught us that an appraisal is synonymous with harsh criticism and a dressing down for underperformance. Such a view is erroneous and a misconception which can be removed by understanding the real purpose of appraisals.

Why appraisals?

Traditionally, performance appraisals have been used by organisations for a variety of different purposes, including salary recommendations, promotion and layoff decisions and training recommendations. In general, 'performance elements tell employees what they have to do and standards tell them how well they have to do it' (United States Department of the Interior, 2004).

Such an overarching definition, however, can render the appraisal process ineffective and even counterproductive to employee performance. One key element that must be emphasised during the appraisal process is that the appraisal is for improvement, not for blame or scathing criticism.

Creating an appropriate appraisal process

One major problem in creating an appraisal process is that no single performance appraisal method is perfect for every organization. The most crucial task in the process is determining proper job dimensions that can be used to gauge the employee against accepted standards that affect the performance of the team or the project.

Peter Drucker developed a method termed 'Management by Objectives' or MBO, in order to address the creation of such job dimensions. Drucker suggested that the objectives of any employee can be validated if they pass the following six tests (Management by Objectives—SMART, 2007):



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- **Specific**
- **Measurable**
- **Achievable**
- **Realistic**
- **Time-related**

If an objective meets these criteria, it is considered a valid dimension on which to gauge performance. The standards on which the objective is compared with should also be validated using the SMART method.

Performing the appraisal

Performing an appraisal on employees can be an anxiety-ridden experience for both parties if the situation is not handled appropriately, and is therefore seen as one of the most difficult tasks that managers need to perform. There are many acts a manager can perform to make the process easier on both parties, and hopefully, mutually beneficial.

Many of us erroneously believe that that performance appraisals are meant to identify weaknesses to be worked on. Needless to say, the appraisal process will be a demotivating exercise if the manager or the team member participates in the exercise with such a mind-set. Therefore, organisations should be leveraging the strengths of each employee rather than focusing on their weaknesses.

The frequency of appraisal is another important factor that has undergone a long process of evolution. Yearly performance reviews are becoming increasingly rare as organisations are beginning to see the benefits of half-yearly and quarterly appraisals.

The *Project Manager* administering the performance appraisal must see it has a two-way conversation rather than a top-to-bottom monologue. Inviting feedback and listening to reactions and concerns from the employee during the appraisal process becomes very important to establishing trust with the employee

If the appraiser provides any negative feedback or improvement points, suggestions should be made to help resolve the problem to develop the *Team Member's* performance. This should be followed by periodic follow-up to assist with any problems with the development and to track progress, rather than waiting until the next performance review

Appraisal can be even more effective and less intimidating if employees have the opportunity to appraise their managers as well as their peers. With this 360-degree feedback process, employees and managers will see the various facets of their performance and will be able to participate on an even playing field, ultimately providing a greater ability to work together to achieve the project or organisational goals.

Performance appraisals should not be looked upon as a necessary evil, but rather a process that has the ability to develop and improve the people within the company. By taking the time to create appropriate performance measures, and administering them accordingly, the resulting system can provide long-term gain for the project/organisation.



One-on-Ones

In the project management lexicon, one-on-ones are meetings between two people involved in the project. These meetings are used to discuss priorities, to resolve issues, brainstorm on key decisions, to communicate expectations and just about anything that can arise in the project that needs to be discussed individually with a team member rather than with the entire team.

During the project kick-off

When a project is initiated, there is more to it than just the formal kickoff with the relevant stakeholders. As we have learned in previous segments, the project kick-off is key and requires a good deal of planning and co-ordination, but it is also important that the *Project Manager* know his team and that he sets expectations with them and makes sure that they understand the goals of the project.

Essentially, the *Project Manager* needs to make sure that each team member has all of the information they need to get started on the project. Many times—depending on the size and complexity of the project and the size of the team—this can be accomplished in a team meeting.

However, in projects that require much specialised resources and have complicated goals or involve risks and requirements that need a more grass roots clarification and discussion, this may not be possible.

In addition, the individual one-on-one meeting between the *Project Manager* and the *Team Member* can be very useful in getting both of them to know each other, which will serve as a foundation for trust and understanding moving forward for two individuals who may be meeting or working together for the first time.

These team member one-on-ones should be used to clarify priorities and to discuss schedules and plans. If some of the resources have worked with the *Project Manager* in other past projects, then some details such as the management style and the preferred communication style—may not need to be discussed.

One-on-Ones during the project life cycle

One-on-ones with key players should also be scheduled on a regular basis throughout the project. For a short project, the one-on-ones might happen daily or weekly with key personnel. For a longer project with more players, formal one-on-ones might be monthly or even less frequent depending on the complexity of the project and the on-going health of the project.

In the first one-on-one meetings, the following should be clarified and reviewed with each individual:

Why he/she was selected: What specific skill sets are needed by the project that the team members have to offer.

Individual priorities, tasks and milestones: This applies to both the sides. What are the priorities, tasks and milestones for them as the *Project Manager* on the project and what does the team member have on his/her plate in terms of other responsibilities.



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Methodology: This basically refers to how one is going to manage the project, the administrative procedures and project management methods and tools that will be used.

Performance expectations: This will include the *Project Manager's* expectations for the *Team Member* in terms of performance and commitment.

Challenges and issues: This will include what the *Project Manager* sees as project challenges and potential individual challenges for the resource.

Processes for solving problems: This is a very important point as the *Project Manager* will want to discuss how decisions will be made, problems will be dealt with and what will be the communication processes for these.

The schedule for future one-on-ones: Finally, an effective one-on-one must set expectations for when these meetings will continue in the future. Ideally, a *Project Manager* will also discuss the possible action items for the next session.

A close-knit, collaborative team that gels together is very important to project success. If such a team atmosphere exists, one-on-ones are not that essential. However, this is an ideal situation and in most real-life projects, invariably, there are bound to be issues and problems that will need to be addressed on a daily basis and a team meeting may not be suited to such discussions and may in fact exhaust everybody's time when only two individuals could have had a productive discussion in a one-to-one.

To conclude, people management is one of the most valuable skills that a *Project Manager* can develop. An effective *Project Manager* will take the time to get to know his *Team Members* and develop a certain level of trust and understanding, which will help him emerge as a natural leader who will invariably get more work accomplished through his team.

Conducting appraisals and one-to-ones are also part of the broader study of people management as both require communication and interaction with people and need a specific skill set and comfort in dealing with people, at large.

Summary

In this segment, you learned that:

- people management is critical for the success of a project and requires the *Project Manager's* utmost attention and dedication,
- an appraisal should be a collaborative exercise where the *Project Manager* and the *Team Member* get together to constructively discuss the strengths and accomplishments of the *Team Member* and the strategies or improvement,
- an appraisal should not be an intimidating exercise where the appraisee is subjected to harsh criticism and rebuke, and
- one-on-ones are individual meetings between the *Project Manager* and the *Team Member* where such issues are discussed, which do not need the team presence.



Meeting the Resource Needs of a Project

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1. Introduction
2. Meeting the Resource Needs of a Project
3. Summary

Objectives

Upon completion of this segment, you will be able to:

- Differentiate between human and non-human resource
- Explain the Project Manager's role in meeting the resource needs of a project
- Identify the factors that need to be considered while finding the appropriate human resources

Introduction

In the last segment, we learned about the importance of people management and the purpose of team appraisals and one-on-ones. In this segment, we will focus upon the process of meeting the resource needs of a project.

Resources are integral elements without which no project can be successfully executed. Resources can be either human or non-human. Non-human resources are materials, machines, equipment and so on, whereas human resources refer to people.

During the initial stage of a project, resources are approximated based on the scope and the expected completion time of the project. The *Project Manager* provides a high-level resource plan to the *Sponsor* and other *Stakeholders* for review and approval.

The *Sponsor* authorises the mobilisation of resources for the project. He/she can delegate and authorise the responsibility of finding the right resources as and when required for the project.

Organisations might also have their own processes for obtaining resources. The *Project Manager* needs to know these processes and draw a plan accordingly.

Loading all resources upfront to a project may not help. This could increase the cost of the project due to upfront stocking. The generic trend is that the resource requirements is initially low in a project and peaks only during execution and again dips at the closure.

Meeting the Resource Needs of a Project

Resources are the backbone of a project. As a project cannot exist without a *Sponsor* and a *Project Manager*, it cannot also not be executed without the appropriate resources.

Resources required in a project can be either human or non-human. Human resources, obviously, refer to the manpower. The reason for using the word non-human is that it could be machines, materials, data or anything which is necessary for the successful completion of a project.



During the project initiation phase, there may be many assumptions regarding the resources. These assumptions have to be carefully analysed and the *Project Manager* has to prepare a proper resource plan. The plan should mention when and how many resources are required. It can provide many details regarding the resources, which may be pivotal for the project.

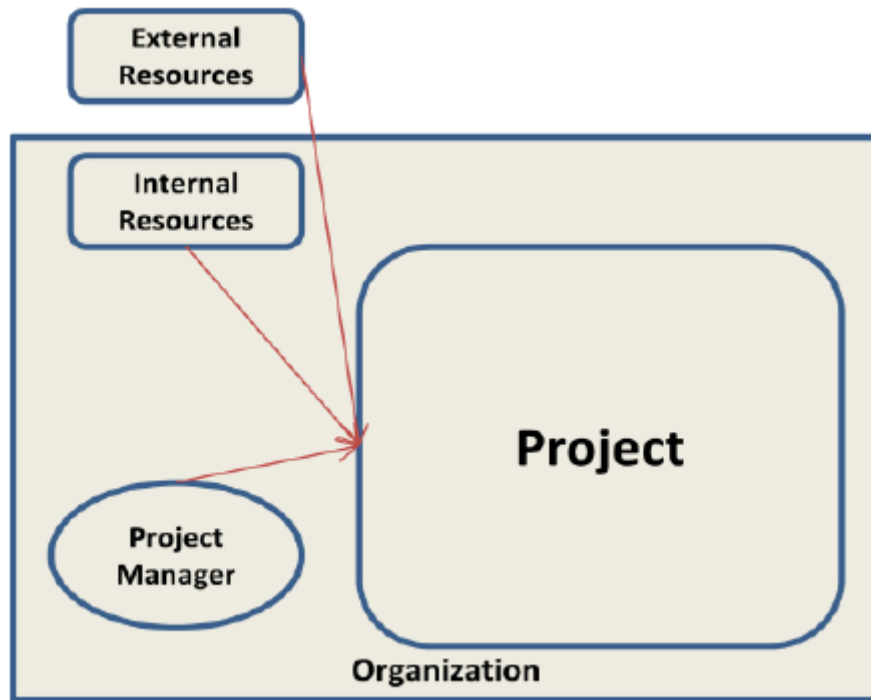


Figure 31: Resources in the Organisational Framework

Are human resources easily available for a project?

It depends on the size and structure of company/organisation. In large IT organisations, human resources may be easily available. Other small organizations may have to hire based on the size and duration of the project. Proper requests have to be sent by the *Project Manager* seeking the resources. The resource plan comes in handy at this stage.

The *Project Manager* in any organisation needs to know the process to find the appropriate resources. He/she needs to ask some pertinent questions at this stage:

- Are there any forms to be filled in and sent?
- Does a request need to be raised through a particular software?
- What is the lead time for getting the resources after raising the request?

The *Project Manager* has to understand these nitty-gritties and work along to accomplish the stated goal.

In the case of manufacturing companies, human resources may have to be loaned from different departments or groups. This can apply to machinery too. There is a high level of interaction (communication) involved in such cases. Proper understanding of processes and policies will help the *Project Manager* to make informed decisions.



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Some of the aspects that need to be examined while finding appropriate human resources for projects are:

- skills required;
- the proficiency level (novice, intermediate, advanced or expert);
- the likely date of resource availability;
- communication skills;
- feedback from the previous project or company; and
- attitude.

In the case of materials and machinery required for the project, it is essential to clearly mention the proper specifications and features of the project and the key dates related to the project. In case the *Project Manager* is looking for specific branded materials or machinery, he/she will need to highlight the same. This would help the procurement and finance team to make the appropriate decisions at the earliest.

Getting the right resources is one of the key requirements for the success of a project.

The *Project Manager* must have a resource plan, which is updated and has been shared with the supporting *Stakeholders*. The *Business Sponsor* must be made aware if there are any critical issues or challenges with respect to meeting the resource needs for the project.

In case the finance team does not approve a certain machinery due to certain constraints, the *Business Sponsor's* intervention or explicit approval could help.

It is also the responsibility of the *Project Manager* to provide the status of resources to all the *Stakeholders*. This would help them to understand the position of the project from the resource point of view.

In this way, meeting the resource needs of a project is a major responsibility of the *Project Manager* and his/her ability to find the appropriate human and non-human resources can go a long way in ensuring the project's success.

Summary

In this segment, you learned that:

- human resources refer to manpower whereas non-human resources refer to machines, materials, data or anything, which is necessary for the successful completion of a project; and
- skills required, the level of proficiency, date of availability, communication skills and feedback received from the previous project are some of the factors that a *Project Manager* needs to consider while finding the appropriate human resources.



Assigning Work to Resources

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2. Assigning Work to the Resources
3. Summary

Objectives

Upon completion of this segment, you will be able to:

- Describe the process of assigning work to the resources
- Identify the key factors that need to be considered while assigning work to the resources

Introduction

In the last segment, we learned about meeting the resource needs of a project. In this segment, we will focus on process of assigning work to the human resources.

The life of a *Project Manager* would be easy if people themselves took the initiative to take up work and then went ahead to complete it without the need for any additional support.

However, in real-life projects, things are very different.

The *Project Manager's* role is multi-faceted and challenging, with a lot of spice in it!

It would be wrong to say that team members do not come forward on their own to take up tasks. This actually happens in the 'agile software development methodology' wherein a list of requirements are kept in front of the team at the start of each work day and the team members pick up the task that they wish to work on.

These tasks are small pieces of work which can be ideally finished in a day. The same cycle is repeated the next day and the team members pick up a new piece of work, which they complete by the end of day.

However, not all projects are executed in this agile fashion. The *Project Manager* has to create a project work plan and share it with the team members to make them aware of the various tasks available to them.

A *Project Manager* can adopt various techniques for assigning work to the team members. If a work plan is created in Microsoft project plan, it can be shared as is or hosted on a website and the team members can look at it and work. In case the plan is created in an Excel format, it could be shared directly with the team members.

In many cases, the *Project Manager* will have group leads or team leads reporting to him. In such cases, the group lead or the team lead will either create his own project plan and share it or obtain the work details for his team from the *Project Manager* and then share it with the team.



Assigning Work to the Resources

Apart from merely sharing the work with the team members in the form of a plan, there are other considerations that a *Project Manager* has to take into account. These include:

Clarifying what needs to be done and why it is important

The background and the context of the task will make it more meaningful for the people working on it. Clarity can be sought by asking the following questions:

- Why is this job important?
- What is the rationale and purpose behind it?
- Are there any particular *Stakeholders* for whom this is especially important and how?

It may help to be explicit about how much this means to the team, to and spell out the impact that success will have as well as the consequences of failure.

Showing the bigger picture will enable them to understand the tasks better, make it more meaningful and will bring out the best in them.

Making the *Project Manager's* expectations clear:

Spell out precisely what the *Project Manager* expects and provide a date for completion. Express confidence in the person's ability to achieve the task even if it is demanding for him/her. Trust and understanding is important.

Describe what a good result will look like, and any particular expectations from the particular resource. The following questions can prove helpful:

- Who is responsible for what and to whom?
- Who will be expected to work alongside whom, and how do their responsibilities inter-lock?

If there is risk of confusion and issues arising due to improper decision-making, it is essential to convey who is responsible for which decisions.

Explain how results will be viewed by the *Stakeholders*:

Sharing the plan will provide understanding about what the kind of results that are to be achieved and by whom. It is also important to make the resources understand how their task results will be viewed by the other *Stakeholders*.

Where applicable, clearly mention who will review their work and how the feedback will be provided. Also clarify the content and nature of feedback that can be expected.

Sharing regular feedback is essential for ensuring quality work and for improving performance. Share the feedback in a constructive way such that the resources take it positively and work on it instead of getting dejected and letting it affect their performance.



Explain the support services available:

Clearly explain the kind of support and services that will be available for the members in the project. It could be transport service, food service, cafeteria availability during the weekend or anything that will be supportive in nature. Doing so will reduce the number of issues that can crop up at a later stage in the project.

Summary

In this segment, you learned that:

- a *Project Manager* can adopt various ways of assigning work to his team, and
- there are various considerations that govern the process of assigning work to the *Project Team*, all of which aim at keeping the team and the various *Stakeholders* informed at all times.



Resource Considerations in a Project

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1. Introduction
2. Resource Allocation
3. Scheduling
4. Project Cost Estimate and Budgets
5. Summary

Objectives

Upon completion of this segment, you will be able to:

- Describe the process of resource allocation
- Describe the process of scheduling
- Describe cost estimates and budgets

Introduction

In the previous segment, we discussed about how the *Project Manager* assigns work to the resources and the various considerations that he/she needs to take into account while doing so. In this segment, we will focus on the various resource considerations in a project including resource allocation, scheduling and project cost estimate and budgets.

After planning and scheduling the project and putting it on the track, it is the resources that must be considered. Resource availability cannot be taken for granted as some may be available in abundance while others may be in scarce supply.

Therefore, the *Project Manager* needs to understand the intricacies of resource balancing with the help of levelling and smoothing tools. Resource levelling, or smoothing, is a technique used to minimise fluctuations in resource utilisation. This method attempts to utilise resources uniformly without extending the project schedule beyond the deadlines.

The resources so required for the project activities should be made available as and when the need arises. These resources include land and building, people, equipment, machines, tools, facilities, etc. The manpower would include both skilled and unskilled labour, executives and other specialist staff such as software developers, assembly line foremen, supervisors and line men.

Resource management adds a vital dimension to the process of planning and scheduling. In many projects, the quantum of various types of resources available to pursue the project activities is limited. Some simultaneous activities would require the same resources while some activities which are performed subsequent to predecessor activities require limited resources.

In other words, there will be heavy usage of a particular (standard) resource during those activities, which are performed at the same time. If sufficient resources are not available during that specified time period, these activities may have to be rescheduled for a later time when resources are available.

Therefore, resource constraints influence the project planning and scheduling scenario.



There are certain activities, also known as critical activities, which deserve maximum attention of the *Project Manager* since any delay in completing these activities will lead to a delay in project completion. The objective of identifying critical activities in a project is to ensure that the project is completed in the shortest possible time by proper monitoring and control of the critical activities. It is the cost consideration that prompts the *Project Manager* to aim at time reduction.

The relationship between cost and time is not that simple. It cannot be said that time reduction will always lead to cost reduction. If a particular activity can be completed at a certain cost in a normal timeframe, the activity may cost less or more when its completion time is shortened. This depends upon the nature of the activity and the resources used for completing the activity.

In this segment, we will first attempt to explain the term resource allocation, and will then focus on the cost aspect of the project.

Resource Allocation

The term resource allocation refers to fixing the available resources to the prioritised project activities. Resource allocation is a part of resource management. In project management, the term resource allocation can be defined as the scheduling of activities and the resources required by those activities while taking into consideration both the resource availability and the project time.

In strategic scheduling, resource allocation is a plan for using the available resources, such as human resources, to achieve goals for the future. It is the systematic way of allocating resources between the various projects or business units. The job of a *Project Manager* is to plan and assign the resources for different activities so that resource utilisation is optimised. Due to resource constraints, a *Project Manager* may face one of the following two situations:

- Resource levelling
- Resource smoothing (fixed resource-limit scheduling)

Resource levelling is used when critical resources need to be shared between different projects in an organisation, or have been over allocated to some activities, which are being performed concurrently within a project, or in other projects in an organisation. Thus, the resource demand exceeds the supply of resources. This technique ensures balance between the demand for resources and their supply.

Resource smoothing is used to maintain the utilisation of project resources at particular limits. For example, if resource A is assigned to a project activity, when the project is being executed in a functional organisation and the resource is available for only 4 hours in a day. Then, resource smoothing technique will be employed to allocate the resources according to the specific limits.

Both, resource levelling and resource smoothing, are employed to optimise the utilisation of resources. However, resource smoothing is applied only on activities that have float, and thus their critical path is not influenced, whereas resource levelling may cause the critical path to be changed. Resource smoothing is only employed for activities that are not on the critical path.

Scheduling

Scheduling the scarce resources is a method for developing the shortest schedule when the number or quantity available is fixed. It is the process of deciding how to arrange resources between a



variety of possible activities and tasks. Time can be particular or floating as part of a sequence of events.

Scheduling significantly drives the project manufacturing and engineering activities and has an impact on the overall productivity. It also aids in the timely completion of a project. The purpose of scheduling is to minimise the production time and reduce the costs. Project scheduling aims to maximise the efficiency of the operation.

Project scheduling tools greatly outperform old manual scheduling methods. The tools provide the project scheduler with graphical interfaces which can be used to optimise real-time workloads in various stages of the project. Project plan recognition allows the software to automatically create scheduling opportunities which may not be apparent without this view into the data.

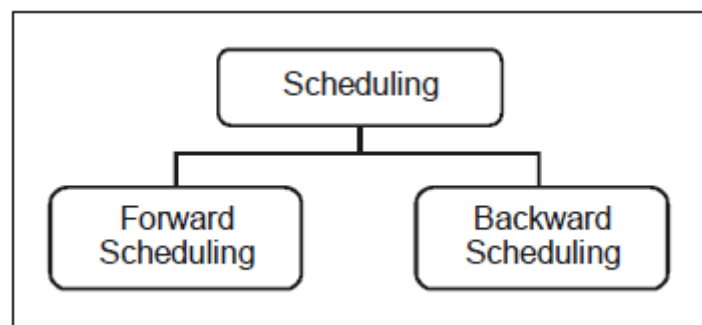


Figure 32: Types of Scheduling

Primarily, companies use backward and forward scheduling to allocate plant and machinery resources, plan human resources, plan production processes and purchase materials. The term forward scheduling can be defined as the planning of tasks from the date the resources become available to determine the shipping date or the due date.

On the other hand, backward scheduling is planning the tasks from the due date or required-by date to determine the start date and/or any changes in the capacity required.

Benefits of Scheduling

The biggest benefit of scheduling is predictability. A scheduling system makes everyone's job easier by adding predictability to the AIS environment. To your superiors, it provides a means of holding down costs through better use of personnel and equipment. Other possible benefits of scheduling areas are:

- process change-over reduction;
- inventory reduction, levelling;
- reduced scheduling effort;
- increased production efficiency;
- labour load levelling;
- accurate delivery date quotes; and
- real-time information.



Project Cost Estimate and Budgets

Budgeted costs of project activities are prepared by estimating scheduled activities and the relative costs of the resources needed to complete each activity. In this process, the *Project Manager* considers the possible causes of variation in the cost estimates including the associated risks.

Cost estimation is the process of identifying and considering different costing alternatives. For example, additional work during the design phase may hold the possibility of reducing the cost of the execution phase and product operations. In the cost estimating process, we consider whether the expected savings can offset the cost of the additional design work.

Cost estimates are expressed in currency units to facilitate comparisons, both, within and across projects. In many countries, the estimator uses man hours or man-days to measure cost factors and facilitate appropriate management control.

Cost estimates can help to refine the course of the project to reflect the additional details. The accuracy of a project approximation will increase as the project progresses through the project life cycle. For example, a project in the initiation phase could encompass a vague idea of estimates in the range of -75 to +100%. Later in the project, when more information is known, estimates could narrow down to a range of -10 to +15%.

In some application areas, there are guidelines to refine and modify the degree of accuracy as per the expectations. The costs for schedule activities are estimated for all resources, which are charged to the project. This comprises, but is not limited to, labour, materials, equipment, services and facilities, with special categories such as an inflation allowance or a contingency cost.

A schedule activity cost estimate is a quantitative evaluation of the expected costs of the resources required to complete the scheduled activity. If the performing organisations do not have formally trained project cost estimators, then the *Project Team* will require supplying both the resources and the knowledge to perform project cost estimating activities.

Cost estimation: Inputs

Enterprise environmental factors

The cost estimating process considers:

Marketplace conditions – Type of products, services and results that are available in the marketplace, and the associated terms and conditions of their availability.

Commercial databases – Information about the resource cost rate is often available from commercial databases that track skills and human resource costs and provide standard costs for the material and equipment. The published seller price lists are another source.

Organisational process assets – In developing the cost management plan, the existing formal and informal cost estimation-related policies, procedures and guidelines (in which cost estimating tools, examining and reporting methods are used) are considered.

Cost estimating policies – A number of organisations have predefined approaches to cost estimation. Where these exist, the project operates within the boundaries defined by these policies.



Cost estimating templates – Several organisations have developed templates (or a pro forma standard) for the use of the *Project Team*. The organisation can constantly improve the template based on its application and usefulness in prior projects.

Historical information – Information (that pertains to the project's product or service) which is obtained from various sources within the organisation can influence the cost of the project.

Project files – One or more of the organisations involved in the project will maintain records of previous project performance that are detailed enough to aid in developing cost estimates. In some application areas, individual team members may maintain such records.

Project team knowledge – Members of the *Project Team* may recall previous actual costs or cost estimates. While such recollections can be useful, they are generally far less reliable than documented performance.

Lessons learned – Lessons learned could include cost estimates obtained from previous projects that are similar in scope and size.

Project scope statement

A project scope statement describes the business need, justification, requirements and current boundaries for the project. It provides important information about project requirements that are considered during cost estimation. The project scope statement includes constraints, assumptions and requirements. Constraints are specific factors that can limit the cost estimation options. One of the most common constraints for many projects is a limited project budget. Other constraints may include required delivery dates, available skilled resources and organisational policies.

Assumptions are factors that will be considered to be true, real or certain. Requirements with contractual and legal implications can include health, safety, security, performance, environment, insurance, intellectual property rights, equal employment opportunity, licences and permits—all of which are considered when developing cost estimates.

The project scope statement also provides the list of deliverables and acceptance criteria for the project and its products, services and results. All factors are considered when developing the project cost estimate. The product scope description, within the project scope statement, provides product and service descriptions and important information about any technical issues or concerns that are considered during cost estimation.

Work Breakdown Structure

The project's Work Breakdown Structure (WBS) provides the relationship among all the components of the project and the project deliverables.

WBS dictionary

The WBS dictionary and related detailed statements of work provide identification of the deliverables and a description of the work in each WBS component required to produce each deliverable.



Project management plan

The project management plan gives the overall plan for executing, monitoring and controlling the project, and comprises subsidiary plans that offer guidance and direction for cost management planning and control.

Schedule management plan

The type and quantity of resources—and the amount of time those resources are applied to complete the work of the project—are a major part of determining the project cost. Schedule activity resources and their respective durations are used as key inputs in this process. Activity resource estimation involves determining the availability and quantities required of staff, equipment and materials needed to perform the scheduled activities. It is closely coordinated with cost estimation.

Activity duration estimation will affect cost estimates on any project where the project budget includes an allowance for the cost of financing, including interest charges, and where resources are applied per unit of time for the duration of the schedule activity.

Schedule activity duration estimates can also affect cost estimates that have time-sensitive costs included in them such as union labour with regularly expiring collective bargaining agreements, materials with seasonal cost variations, or cost estimates with time-related costs such as time-related field overhead costs during the construction of a project.

Staffing management plan

Project staffing attributes and personnel rates are necessary components for developing the schedule cost estimates.

Risk register

The cost estimator considers information on risk responses when producing cost estimates. Risks can be either threats or opportunities. They usually have an impact on, both, the schedule activity and project costs. As a general rule, when the project experiences a risky event, the cost of the project will always increase, and there is likelihood of delay in the scheduled project.

Cost estimation: tools and techniques

Analogous estimation

Analogous cost estimation implies using the actual cost of previous (similar) projects as the basis for estimating the cost of the current project. Analogous cost estimation is frequently used to estimate costs when there is a limited amount of detailed information about the project (For example, in the early phases). Analogous cost estimation uses expert judgment.

Analogous cost estimation is generally less costly than the other techniques but is generally less accurate. It is most reliable when the previous projects are similar in fact and not just in appearance, and the persons or groups preparing the estimates have the needed expertise.



Determination of resource cost rates

The person determining the rates or the group preparing the estimates must know the unit cost rates such as staff cost per hour and bulk material cost per cubic yard for each resource to estimate the schedule activity costs.

Gathering quotes is one method of obtaining rates. For products, services or results that are to be obtained under contract, standard rates with escalation factors can be included in the contract. Obtaining data from commercial databases and seller-published price lists is another source of cost rates. If the actual rates are not known then the rates themselves will have to be estimated.

Bottom-up estimation

This technique involves estimating the cost of individual work packages or individual schedule activities with the lowest level of detail. This detailed cost is then summarised or 'rolled up' higher for reporting and tracking purposes. The cost and accuracy of bottom-up cost estimation is typically motivated by the size and complexity of the individual schedule activity or work package. Generally, activities with smaller associated effort increase the accuracy of the schedule activity cost estimates.

Parametric estimation

It is a technique that makes use of a statistical relationship between historical data and other variables (for example, square footage in construction, lines of code in software development and the requisite labour hours) to compute a cost estimate for a schedule activity resource. This technique can generate higher levels of accuracy depending upon the sophistication, the underlying resource quantity and cost data built into the model. A cost-related example consists of multiplying the planned quantity of work to be executed by the historical cost per unit to obtain the estimated cost.

Project management software

This includes cost estimating software applications, computerised spreadsheets, and simulation and statistical tools that are extensively used to assist with cost estimation. Such tools make it simpler to estimate cost and thus facilitate the consideration of various cost estimate alternatives.

Vendor bid analysis

Vendor bid analysis is the analysis of what the project should cost. In the case of problems where projects are won under competitive processes, additional cost estimation may be required from the *Project Team* to scan the price of individual deliverables in order to derive a cost that supports the total project cost.

Reserve analysis

In many schedule activity cost estimates, various cost estimators include reserves (contingency allowances) as costs. This may lead to potentially overstating the cost estimate for the schedule activity. Contingency reserves are estimated costs to be employed at the discretion of the *Project Manager* to deal with probable, but not certain, events. These events are 'known unknowns' and are an element of project scope and cost baselines.



One alternative to manage cost contingency reserves is to aggregate each scheduled activity's cost contingency reserve—for a group of correlated activities—into a single contingency. This scheduled activity may be a zero period/duration activity that is placed in the entire network path for that group of scheduled activities and is employed to hold the cost contingency reserve.

An example of this solution to manage cost contingency reserves is to assign them at the work package level to a zero duration activity which spans from the start to the end of the work package sub-network.

As the schedule activities progress, the contingency reserve, as measured by resource consumption of the non-zero duration schedule activities, can be adjusted. As a result, the activity cost variances for the related group of schedule activities are more accurate because they are based on cost estimates that are not pessimistic. Alternatively, the schedule activity may be a buffer activity in the critical chain method and is intentionally placed directly at the end of the network path for that group of schedule activities.

Cost of quality

Cost of quality can also be used to prepare the schedule activity cost estimate.

Cost estimation: Outputs

Activity cost estimates

An activity cost estimate is a quantitative assessment of the likely costs of the resources required to complete the schedule activities. This type of an estimate can be presented in a summary form or in detail. Costs are estimated for all resources that are applied to the activity cost estimate. This includes, but is not limited to, labour, materials, equipment, services, facilities, information technology and special categories such as an inflation allowance or cost contingency reserve.

Activity cost estimate supporting detail

The amount and type of additional details supporting the schedule activity cost estimate vary according to the application area. Regardless of the level of detail, the supporting documentation should provide a clear, professional and complete picture by which the cost estimate was derived.

The supporting details for the activity cost estimates should include:

- explanation of the schedule activity's project scope of work;
- documentation of the basis for the estimate (that is, how it was developed);
- documentation of any assumptions made;
- documentation of any constraints; and
- signal of the range of possible estimates (For example, USD 10,000 (-10%/+15%) to indicate that the item is expected to cost between USD 9,000 and USD 11,500).

Requested changes

The cost estimating process may generate requested changes that may affect the cost management plan, activity resource requirements and other components of the project management plan. Requested changes are processed for review and disposition through the integrated change control process.



Cost management plan (updates)

If approved change requests result from the cost estimating process, then the cost management plan component of the project management plan is updated if those approved changes impact the management of costs.

Summary

In this segment, you learned that:

- resource allocation is a part of resource management and refers to fixing the available resources to the prioritised project activities;
- scheduling is the process of deciding how to arrange resources between a variety of possible activities and tasks;
- cost estimation is the process of identifying and considering different costing alternatives; and
- there are various tools and techniques available for cost estimation, which include analogous estimation, determination of resource cost rates, bottom-up estimation and parametric estimation.



Managing Resources

Table of Contents

1. Introduction
2. Managing Resources
3. Challenges Posed by Organisational Structures
4. Managing Conflict
5. Summary

Objectives

Upon completion of this segment, you will be able to:

- Describe how resources are managed in a project
- Identify the challenges posed by organisational structures in managing resources
- Explain how conflict is managed in a project

Introduction

In the previous segment, we learned about the various resource considerations that are necessary in a project. In this segment, we will focus on the concept of resource management and the various challenges faced while managing resources.

Managing resources is frequently understood as managing human resources, that is, people. However, resource management goes beyond the management of human resources and also includes non-human resources which are necessary for the successful completion of the project. Therefore, during the life cycle of a project, the *Project Manager* not only needs to manage his team members but also the equipment used by the project and the materials needed by the people.

Managing Resources

Managing non-human resources

A Project Manager has to manage many kinds of equipment related to the project. For example, cranes, trucks, lathes, computers, telephones and similar other equipment related to the project. While the *Project Manager* is not necessarily required to directly handle and fix them, he needs to have inventory of these items along with the necessary details such as:

- rental/purchase details,
- service warranty or service due details,
- working or non-working status,
- quick solutions for commonly-faced issues,
- contact details for support,
- backup options in case of failure, and
- bus/shuttle route numbers and timings for common locations.

Collating such information and keeping it handy will help the *Project Manager* to provide the required support to the team.



Project Management

The *Project Manager* can share such information with the team members in various ways which include:

- posting it on the project Website,
- displaying it in the common workplace or on the notice board,
- displaying it on equipment or materials for getting quick support—for instance, displaying support desk number on the desktop, telephone or the printer, and
- sending revised or updated information to all team members on their email ids.

Sharing such information will help the team members to be less dependent on the *Project Manager* and will improve productivity and utilisation.

The *Project Manager* may use his support team for assistance to manage such items and have regular meetings to discuss the working status of all items. Even during regular meetings with the team members, questions related to the status of these items can be taken up and noted.

For the *Project Team Members* who need to often travel out of city, providing information about accommodations, safe routes to reach and so on will help the *Project Manager* in ensuring that the team members are comfortable while completing their work.

Managing human resources

Managing human resources is a critical and challenging job for a *Project Manager*. In fact it is a very tricky one too.

It is easier for one born as a leader but not everybody is a born leader.

Imagine that it is a beautiful winter evening and you are enjoying a cup of tea with your family and friends. You are meeting some of them after a long time and are eager to catch up on old times

Just then, your phone rings. It is from your *Project Team Member*. Your family and friends would rather that you did not answer the call. But, you are a *Project Manager*! You have to. You take the call hoping that it is nothing critical and will intrude in your private time for very long.

It is your team member who is working on your project. You are aware that they are working hard not just late evenings but weekends too as they are behind schedule. They often need your guidance, approval and attention to move ahead and patch up that schedule gap.

But you have to be there with your team now.

How do you feel?

This is not the only instance you will face as a *Project Manager*. As you gear up for being a *Project Manager* you will have to face many challenges like such as slipping work, lack of resources, incompetence of certain team members, not having the right equipment to complete the job, lack of training, de-motivated team members and too many changes in the project.

Challenges will be aplenty but not all of them will come up in one project. It will depend on many factors and also on how you start and nurture the project from the beginning.



Project Management

Everything might have looked perfect on paper when you planned it. However, when it comes to execution, there will be many constraints related to human resources. The biggest influence on the *Project Manager's* control of the *Project Team* is the organisational structure.

In individual projects, there will be different kinds of constraints. However, when it comes to the execution of a project in a company/organisation, the structure of the organisation plays an important role.

While we have discussed the organisational structures in a previous segment, let us glance over these different structures once again—specifically, in the context of how they influence the *Project Manager's* ability to manage resources.

Functional structure: Here, the *Project Manager* will have little or no power. The organisation will be grouped on the basis of functions such as marketing, sales, IT and so on. The functional manager of each group will have more power. If your project lies within one function, and you have the complete authority to manage resources, then it would be easy. However, if you are from a different function and have to use resources only for work, it would then be a challenge.

Weak matrix structure: In this type of a structure, the *Project Manager* will have a little more power but still less than the functional manager. Here resources are used from across the organisational groups. Team members will be on multiple projects at the same time and it can become chaotic. There can be many questions: What are the deadlines of each project? Who is in charge? Which project is a priority? In this structure, the *Project Manager* is still a coordinator and has to communicate with all groups and members.

Balanced matrix structure: This model is a bit mature as compared to the weak matrix where the *Project Manager* and the functional manager have equal powers. This is where it gets tricky as such a situation is often fraught with a power struggle where both managers end up competing against each other to demonstrate their powers. There is likely to be a lot of politics and power struggle. Team members would still be drawn from across the functional groups but the power divide could dilute the project goal.

Projectised structure: This organisational structure provides more power to the *Project Manager* and there is a team, which is dedicated to his/her project and reports directly to him/her. The *Project Manager* and the team members are happy and comfortable in such a structure. Communication demands are reduced. The *Project Manager* is the focal point and has better control on the project and team.

Composite structure: This is an organisational structure which is a mix and match of the functional as well as the projectised matrix. Here, the *Project Manager* has an independent team and the functional team provides full support to the project. Many large organisations run composite structures.

Is a proper organisational structure enough to ensure the success of a project? The famous Project Management Institute as part of Project Management Professional has enough material which explains tools and techniques of human resource management. It explains how to acquire, initiate, develop and manage a project team.

Using any of the standards, tools or techniques will help a *Project Manager* to understand the resource needs and how to manage them effectively.



Project Management

The purpose of effective resource management is to ensure that the project is successful. For this, the resources have to be motivated, skilled, productive and efficient.

Conflict management

While working with human resources, one of the challenging areas for the *Project Manager* is to deal with human conflicts. Conflicts between teams or members can arise due to various reasons such as ego, working style, nepotism, unhappiness with work and so on.

The *Project Manager* has to build skills to manage these conflicts efficiently. An effective skill that comes in handy here is listening. In order to resolve conflicts, the *Project Manager* has to act as a mediator, coach and a guide. Not all conflicts, however, can be resolved. If the *Project Manager* can resolve conflicts such that both parties have a win-win situation, it is great. However, it is tough to achieve this all the time. Nevertheless, agreeable solutions have to be achieved so that focus is not shifted from the project goal and all parties agree to work together.

Managing risks

We discussed about managing risks in an earlier segment. Risks are identified in the early stage of a project. However, they do not end there. Risks can come in front of you at any point in the project. While executing the projects, the *Project Manager's* mind, eyes and ears have to be open to spot the risks.

Machines not working as planned, team members taking leaves very often or leaving office early during critical moments, weather changes, announcement of strike in the city and many such issues could be risks to the project.

Asking open-ended questions during status meetings with the team members will help the *Project Manager* to identify risks and issues. An effective Project Manager will look out for the following key phrases:

- 'Little or some more time required for work'
- 'Unable to solve immediately'
- 'Waiting for a response'
- 'No confirmation yet from the vendor, client, partner, etc.'

The above issues are only samples. The *Project Manager* or the *Lead* working closely with the team can also look for non-verbal cues by observing how the team members respond through their body language and tone of voice.

An alert *Project Manager* will probe even more and identify if the team is facing risks in its day-to-day activities. For many of these, the *Project Manager* can provide immediate solutions and if he sees that the impact of certain risks is high, then he can log it in the risk document and track it closely in formal risk-status meetings.

Some of the risks perceived on the ground might have already become issues. The *Project Manager* needs to work intelligently to resolve them. This is where, the *Project Manager's* attitude, ability to act quickly and relationships within the organisation help to provide solutions for day-to-day issues. He/she should be aware of the organisational structure and be ready to obtain help for his team which may be in terms of additional resources, technical support from experienced people or direction from senior leads.



Project Management

Projects fail due to inattention to basic control principles of project management. Too many times, the *Project Team* is busy getting on with 'completing the project' and not spending time and energy in anticipating problems. Once a problem is suspected, the team acts too slowly to resolve or even get into the root of the problem.

Preventing problems is far easier and less costly than solving them and the best way to identify a problem is to always look for it. For example, frequent and unchecked changes to the requirements specifications are a known and leading source of design-related problems. Some of the major control issues include:

- poor time and cost estimates,
- frequent and uncontrolled changes to the baseline requirements,
- difficulties in concluding the project because of lack of a completion criteria,
- frequent replacement of development personnel, and
- inadequate tracking and directing of project activities.

Once a project has started, one of the most common problems is that the *Project Manager*, and possibly the full project team, is unaware of the existence of a major problem at a stage when it could be contained and corrected. This can be improved by the consistent sharing of information and taking action based on that information.

The tracking, monitoring and reviewing processes apply in varying degrees in all projects. As in all the sections of this document, care should be taken not to use the processes as a 'recipe,' but to analyse the specific project and develop management structures that best meet the needs of the particular project.

Summary

In this segment, you learned that:

- in the context of project management, resource management refers to the management of both, human as well as non-human resources;
- one of the factors that affect the *Project Manager's* ability to manage resources is the organisational structure;
- managing conflict and risks is one of the major responsibilities of the *Project Manager* during resource management.



Motivating the Team Members

Table of Contents

1. Introduction
2. Motivation
3. Strategies for Motivating a Team
4. Summary

Objectives

Upon completion of this segment, you will be able to:

- Explain the concept of motivation
- Identify strategies for motivating a team

Introduction

In the previous segment, we learned about managing resources and the factors that influence a *Project Manager's* ability to do so. In this segment, we will focus on the importance of motivation and some of the ways in which a *Project Manager* can motivate his/her team members.

Motivation

Do you want to learn how to motivate someone?

Well, you cannot!

That's right—you cannot motivate another person.

Motivation is a personal choice. The only person you can directly motivate is yourself!

However, this does not mean that the only kind of motivation is self-motivation. You can certainly create the '*opportunity*' for other people to become motivated, but you cannot make the decision for them. The same applies to a *Project Manager* who wants to motivate his/her team. He/she cannot directly motivate the team but can certainly create the '*opportunity*' for them to become motivated.

Following are the factors that encourage a person to become and remain motivated to achieve a goal:

- Desirability: Is the goal valuable or desirable enough to take the trouble?
- Feasibility: Is there a likelihood that I can achieve the goal?
- Progress: Am I making significant accomplishments on my way to achieving the goal?
- Reward: Is there a worthy payoff on the achievement of the goal?

When a project meets people's professional and personal needs in each of these four areas, the *Team Members* of such a project are motivated to make the project a success.



Strategies for Motivating a Team

There are many strategies for motivating a team. These vary according to the cultural influences, the Project Manager's leadership style, the organisational culture, the team composition, the team members' openness to try out new ideas and their receptivity to suggestions.

There are several other reasons which could influence the choice of strategy when motivating a team. However, there are some universal strategies which work across cultures and countries and can be applied to most projects.

Following are some such strategies:

Increasing commitment by communicating the project's benefits

While a team will commit to completing a project because their *Project Manager* has asked them to do and they are being paid to do the job, they will be much more committed to the project if they could recognise and appreciate the benefits of that project.

Therefore, a *Project Manager* must be able to communicate the benefits of the project that are most important to the organisation, its employees, and its clients. This will ensure that the team members have a better and first-hand understanding of the benefits of the project.

Encouraging persistence by demonstrating project feasibility

A project is *feasible* if it is possible to be accomplished. If the team members are not convinced that a project is achievable, no matter what the *Project Manager* does, the project will remain on a collision course and will eventually prove to be a failure.

The *Team Members* will be easily discouraged when they encounter the slightest of difficulties which, in turn, will slow down their pace of work and affect their productivity. Therefore, the *Project Manager* must ensure that the *Team Members* are assured of the feasibility of the project and have no doubts about the fact that the project is achievable.

The *Team Members* do not need a guarantee of success but they must know that they have a reasonable chance of success provided they put in efforts. If a *Project Manager* is able to do that, he would have ensured that his team continues to put in the required efforts and does not give up in the face of temporary failures.

Letting people know how they are doing

Getting the team members to appreciate the project's benefits and feasibility helps the *Project Manager* motivate the team initially. However, if the project lasts longer than a few weeks, the team's initial motivation can die out without continual reinforcement.

Therefore, it is important for the *Project Manager* to sustain the initial enthusiasm of the team by providing them feedback at regular intervals (so that *team members* are aware of how they are performing), pointing in the right direction (so that *team members* are able to find their own answers)



Project Management

and providing guidance (so that *team members* do not get overwhelmed at any given point in a project).

Providing rewards for the work well done

Rewarding people at a project's conclusion for their effort and accomplishments confirms to them that they accomplished the desired results and met the audience's needs. It is a validation of their efforts and an acknowledgement of the fact that the stated objective has been achieved.

It also reassures the *Team Members* that managers recognise and appreciate their contributions. This recognition, in turn, makes it more likely that the team will welcome the opportunity to participate in future projects.

In this way, *Project Managers* can apply many strategies to provide the opportunity to their team members to become motivated and stay that way throughout the life cycle of a project.

An effective *Project Manager* is the one who knows which strategy to employ in a given situation to keep up the morale and motivation levels of his team members.

Summary

In this segment, you learned that:

- motivation is a personal choice and the *Project Manager's* role is to provide the opportunity to the *Team Members* to exercise this choice;
- desirability, feasibility, progress and reward and the factors that encourage motivation; and
- there are various strategies that a *Project Manager* can adopt to encourage his/her team, including communicating the project benefits, demonstrating project feasibility, providing feedback and rewarding performance.